

Large-Scale Synthesis of High-Loading Single Metallic Atom Catalysts by a Metal Coordination Route

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Single atom catalyst (SAC) is one of the most efficient and versatile catalysts with well-defined active sites. However, its facile and large-scale preparation, the prerequisite of industrial applications, has been very challenging. This dilemma originates from the Gibbs–Thomson effect, which renders it rather difficult to achieve high single atom loading ($< 3 \text{ mol}\%$). Further, most synthesizing procedures are quite complex, resulting in significant mass loss and thus low yields. Herein, a novel metal coordination route is developed to address these issues simultaneously, which is realized owing to the rapid complexation between ligands (e.g., biuret) and metal ions in aqueous solutions and subsequent in situ polymerization of the formed complexes to yield SACs. The whole preparation process involves only one heating step operated in air without any special protecting atmospheres, showing general applicability for diverse transition metals. Take Cu SAC for an example, a record yield of up to 3.565 kg in one pot and an ultrahigh metal loading 16.03 mol% on carbon nitride (Cu/CN) are approached. The as-prepared SACs are demonstrated to possess high activity, outstanding selectivity, and robust cyclicity for CO_2 photoreduction to HCOOH . This research explores a robust route toward cost-effective, massive production of SACs for potential industrial applications.

1. Introduction

Single atom catalyst (SAC) is one of the most fundamental and important catalytic materials due to their conspicuous advantages of well-defined active sites,^[1–3] exceptional activity, high selectivity, and stability for numerous important reactions in chemistry.^[4,5] In the past decade, a variety of synthetic methods^[6–11] have been developed to synthesize SACs of alkaline-earth^[8] and transition metals.^[12–19] However, in most approaches a lot of organic solvents must be utilized to sequester single atoms due to their extreme instability in the free-standing atomic state.^[5,20] In order to remove the organic solvents and ultimately obtain target SACs, multiple-step washing or heating is required. The complex procedures would result in significant mass loss and thus a low one-batch yield of SACs, rendering large-scale preparation very challenging.

The catalytic performance of a SAC depends on the amount of active metal

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sites. High loading density of dispersive single atoms ensures a greatly boosted utilization efficiency of metals, providing more active centers that are readily accessible to reactants.^[21] However, the metal loadings in reported SACs to date are usually very low (< 3 mol%).^[4,22,23] Further increasing metal loading leads to the formation of clusters or particles due to the Gibbs–Thomson effect,^[24] deteriorating catalytic performance significantly. Hence, the development of advanced synthesis techniques to achieve ultrahigh-density SACs remains a great challenge. Further, most existing preparation strategies suffer from poor applicability as they are unable to be easily extended to the synthesis of SACs containing other metals. These issues in the synthesis of SACs have severely hampered their industrial application.

Aiming at future widespread application of SACs, great efforts have been devoted to address the above-mentioned critical issues, and there have emerged several prominent strategies that could respectively meet a certain single challenge.^[8,19,25–28] According to the preparation procedures, they can be classified into two categories: pre- and postsynthetic introduction of metals on supports. The former helps to stabilize atomically dispersed metal species and achieve high-mass-loading SACs, while the latter is highly amenable to scale-up. However, the two categories are difficult to be compatible with each other, rendering it rather challenging to overcome all issues in one batch. Moreover, special protecting atmospheres (e.g., argon or helium) are usually required as single metal atoms are prone to oxidation in air.^[4] There still exists a huge gap between laboratory-based and industry-scale syntheses of SACs. It is imperative to achieve the large-scale synthesis of SACs with high loading density in air toward the industrialization of SACs.^[24,29]

The metal coordination chemistry, which has a long history of over 150 years, is an important topic in the field of chemistry and has revolutionized the manufactures of many vital industrial and domestic products,^[30] spawning up to eight Nobel Prizes.^[31] Specifically, a half of them were awarded for the pioneering contributions in the field of catalysis and the developed catalysts are all metal coordination complexes. Such complexes commonly consist of a central metal atom species and a surrounding array of ligands.^[32] The local structure of the metal center resembles closely that of SAC. Additionally, in the formation process of the local structure, there are two common characteristics: i) the sources of metal species are usually ions and thus positively charged, leading to the sequestration of single metal species and the formation of tightly bonded metal–ligand units; ii) subsequently these elementary units directly polymerize to form metal coordination complexes without the necessity of removing ligands.

Inspired by the knowledge of metal coordination complexes, we propose a novel and robust metal coordination strategy to synthesize SACs. Taking the SAC of Cu single atoms on carbon nitrile (Cu/CN) as an example, our method achieves a record high one-batch production of 3.565 kg and a high metal loading of 16.03 mol% (i.e., 9.97 wt%). The as-prepared Cu/CN exhibits an outstanding performance toward photocatalytic CO₂ reduction to HCOOH. Furthermore, this strategy is applicable for preparing ten types of M/CN SACs (M = Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, and Zn), and the whole synthetic procedures are in air. Thus, our study promotes a solid step forward for an easy operational and

feasible industrial route of manufacturing SACs and CO₂ reduction in low cost.

2. Results and Discussion

2.1. Synthesis of Large-Scale and High-Metal-Mass-Loading SACs

Figure 1a outlines the whole synthetic procedures of our proposed strategy. Specifically, the process starts at the synthesis of biuret from thiourea by simple ultrasonic treatment in an aqueous solution (Figure 1a1,a2). Upon introducing metal chlorides, a rapid complexation between the metal ions and ligands (e.g., biuret) occurs at room temperature via stirring to yield the metal–ligand complexes (Figure 1a3,a4). Subsequently, a single heating step is required for the polymerization of the resultant complexes to produce SACs (Figure 1a5), during which triazine rings, melams and melems are successively formed (Figure 1a6–a8). In the whole process, no organic solvent was utilized, and thus multiple-step washing was avoided. Specifically, the complexation maximizes the probability of bonding the metal cations to all available coordination sites of the ligands, and ultimately the SACs is in situ transformed from the formed complexes. As a consequence, aggregation of the metal atoms can be effectively prevented in subsequent polymerization, achieving a high-density loading, while the in situ transformation guarantees the preservation of a large mass amount and in turn the realization of a high mass yield.

Instructed by the above strategy and due to the wide utilization of Cu in numerous reactions,^[7,14,15,33] we designed a series of enlarged experiments for Cu/CN preparation. By augmenting the amount of thiourea from 10 g to 14 kg, the yield of Cu/CN in one pot remain nearly constant and the quantities of the samples increase from 1.8 g to 3.565 kg, as shown in Figure 1b; and Figure S1, Supporting Information. Moreover, the structure, valence state of Cu single atoms, morphology and surface areas of the small-scale and large-scale samples are quite similar to each other (Figures S2–S5 and Table S1, Supporting Information), indicating that our proposed strategy is readily amenable to scale-up.

High mass loading of SACs is another important criterion for the highly efficient and stable catalytic processes. Thus, we gradually increased the synthetic metal loading of Cu/CN from 0.5 to 12.5 wt%. The inductively coupled plasma optical emission spectrometer (ICP-OES) analysis of these samples reveals that the losses of Cu loads are negligible, within 2% (Figure S6 and Table S2, Supporting Information), indicating that the mass of metal can be well preserved. Corresponding aberration corrected high-angle annular dark-field scanning transmission electron microscopy (AC HAADF-STEM) images show that Cu atoms remain dispersive up to the Cu loading rate of ≈9.97 wt% (i.e., ≈16.03 mol%) (Figures S7 and S8, Supporting Information).

Figure 1c shows the yield and mass loading of SACs based on our work and literatures with more details summarized in Table S3, Supporting Information, manifesting the ultrahigh production efficiency of our metal coordination route. The loading is also the record high among currently carbon-based and metal oxide substrates. These results indicate that metal coordination route enables large-scale preparation of SACs and high-density

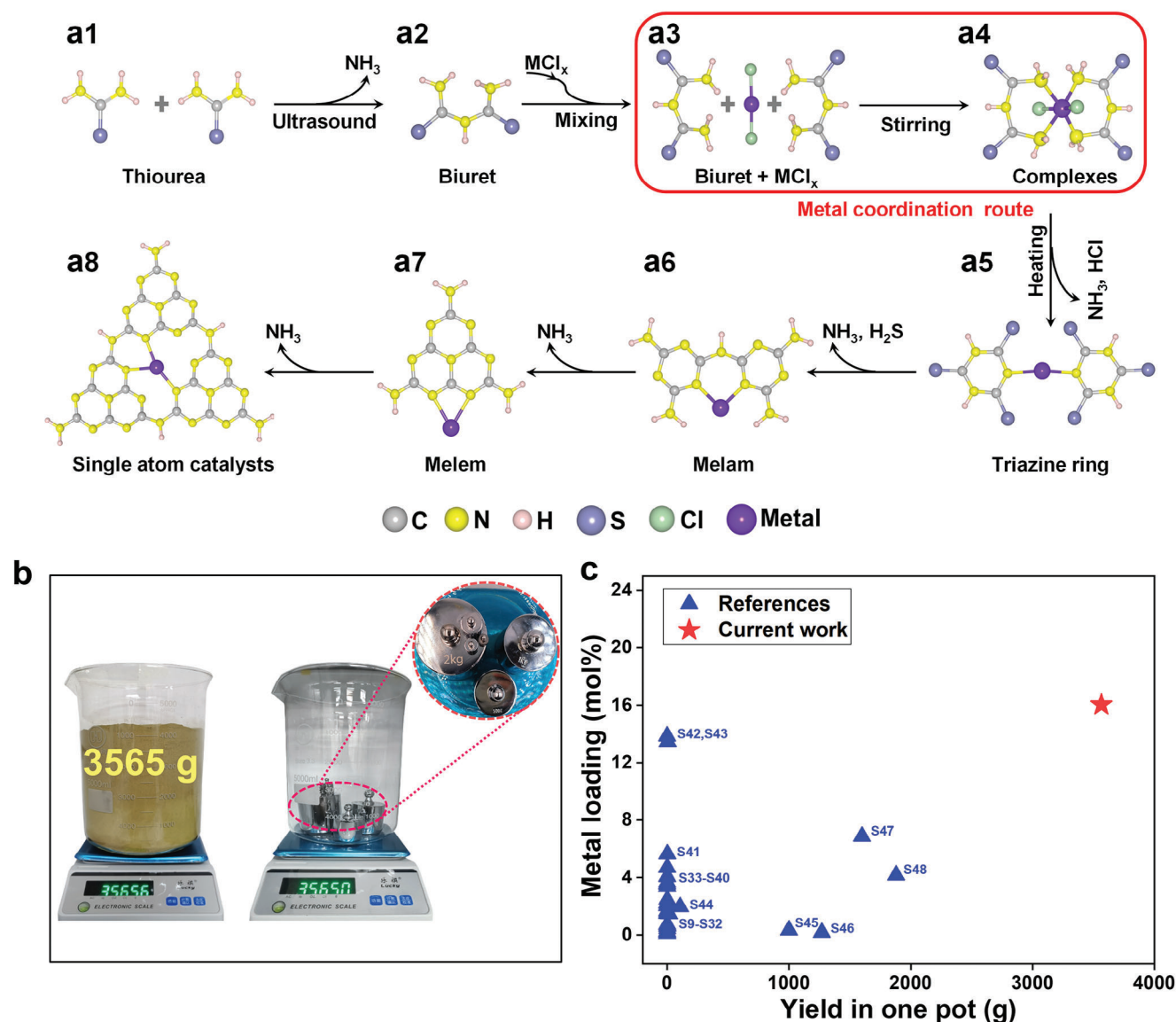


Figure 1. a) The whole synthesis procedures: a1) Thiourea. a2) Biuret. a3–a4) Metal coordination route. a5–a7) In situ polymerization. a8) M/CN SACs. b) Photograph (upper panel) of Cu/CN SAC at a one-pot yield of 3.565 kg. c) Summary of yield of SACs and metal loading (mol%) in our work (red star) and literatures. All references can be found in Table S3, Supporting Information. Gray, yellow, pink, dark blue, purple, and green balls represent C, N, H, S, Cu, and Cl atoms, respectively.

loading, holding potential to meet the requirements of industrial applications. Besides, it is found that copper salts with small anions, such as nitrate and sulfate, can be also utilized as precursors to synthesize Cu/CN SACs (Figures S9–S14, Supporting Information), further confirming the robustness and compatibility of our proposed synthesis strategy.

The crystal structures of the as-obtained Cu/CN SACs products are examined by X-ray diffraction (XRD). As can be seen in Figure S15, Supporting Information, all the XRD patterns of Cu/CN- x (x represents that the Cu loading rate is x wt%; $x = 0.5, 2.5, 5.0, 7.5, 10.0$, and 12.5) samples present two peaks at around 13° and 27° , which are the typical diffraction peaks of CN (JCPDS 87-1526). The former peak is attributed to the in-plane structural packing motif (100), and the latter originates from the interplanar

stacking of (002).^[34] No peaks corresponding to Cu compounds can be detected in the XRD patterns of Cu/CN- x , suggesting the high dispersion of Cu species. Surface area and X-ray photoelectron spectroscopy (XPS) characterizations reveal that the specific surface areas and Cu valence state of the Cu/CN- x samples remain nearly unchanged (Figure S16 and Table S4, Supporting Information). Besides, the S 2p XPS spectra (Figure S17, Supporting Information) indicate that there are no signals for possible S peaks in the range from 160 to 175 eV,^[35] suggesting that there is no residual S in the final product of Cu SACs sample. Moreover, AC HAADF-STEM is carried out to investigate the dispersion state of Cu atoms in the Cu/CN-10.0 sample with the yield of 3.565 kg, as shown in Figure 2a. Individual Cu atoms are clearly visible and evenly dispersed throughout

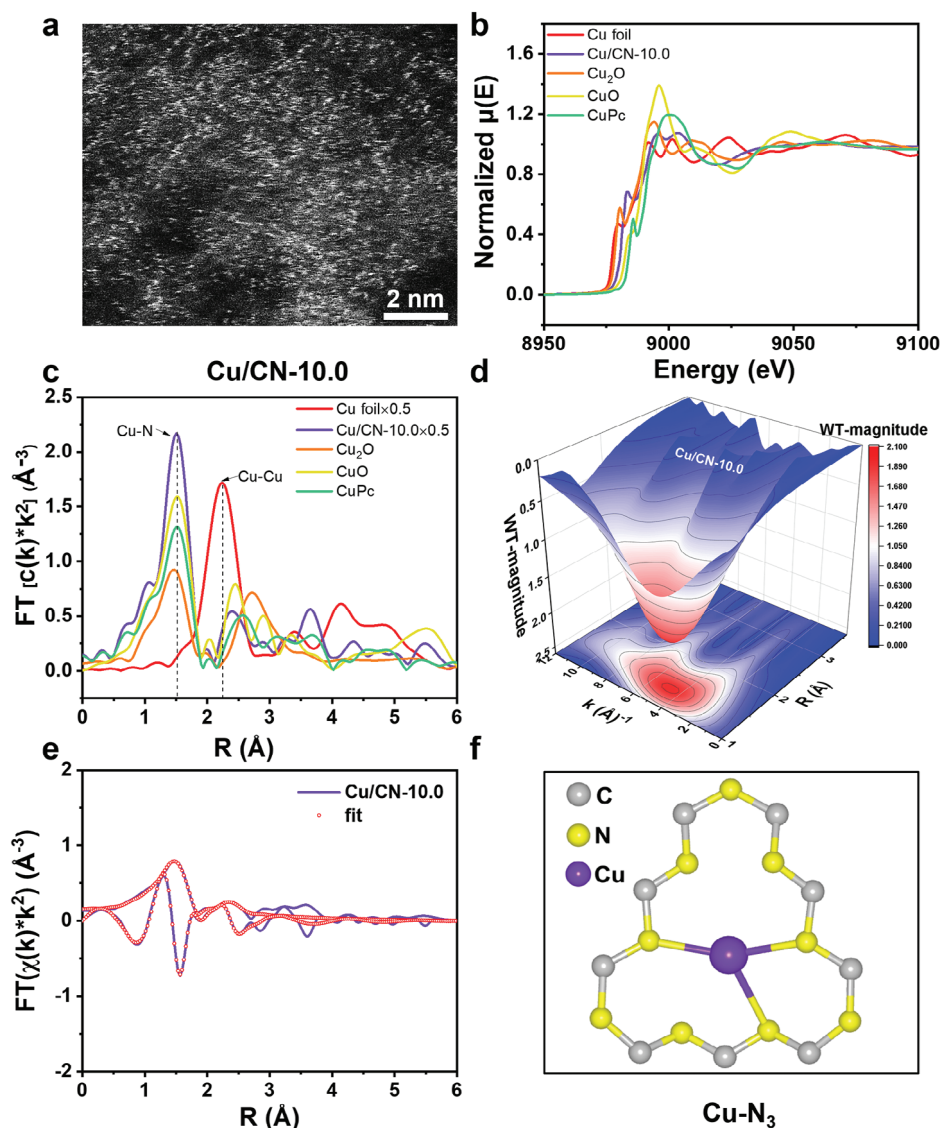


Figure 2. a) AC HAADF-STEM image of Cu/CN-10.0. b) Cu K-edge XANES spectra, and c) FT k^2 -weighted EXAFS spectra of Cu/CN-10.0. d) 3D contour wavelet transforms extended X-ray absorption fine structure (WTEXAFS) maps with 2D projection included Cu/CN-10.0. e) FT-EXAFS fitting curves of Cu K-edge at R space. f) Proposed local structure of Cu single atoms in Cu/CN-10.0, the metal atoms existed as isolated $\text{Cu}^{\delta+}$ centers in a Cu-N_3 coordination environment. Gray, yellow, and purple balls represent C, N, and Cu atoms, respectively.

the entire CN substrate without any metallic Cu nanoparticles or clusters.

2.2. Structural Characterization of the Coordination Environment in Cu/CN SAC

The electronic structures and coordination environments of (Figures S18–S21 and Table S5, Supporting Information) and Cu/CN-10.0 (Figure 2 and Table S6, Supporting Information) are further systematically investigated by X-ray absorption spectroscopy (XAS). The Cu K-edge X-ray absorption near-edge structure (XANES) spectra reveals that both the absorption edge and the main transition energies of Cu in Cu/CN-10.0 are located between those of Cu foil and those of copper phthalocyanine

(CuPc), respectively, suggesting that the Cu atoms carry a partial positive charge in the joint Cu/CN-10.0 structure.^[15] The Fourier-transformed extended X-ray absorption fine structure (FT-EXAFS) spectra (Figure 2c) exhibit a major peak at 1.51 Å. This is larger than that of Cu–O bond (1.46 Å) of Cu_2O , and is consistent with the peak of Cu–N bond in CuPc. Meanwhile, the intensity of Cu–N peak in Cu/CN-10.0 is lower than that in CuPc. Of note, the Cu–N coordination number in CuPc is definitely 4. Thus, this result suggests that the Cu–N coordination in Cu/CN-10.0 is less than 4. Particularly, there are obviously no peaks around either 2.24 Å (corresponding to the Cu–Cu bond) in Cu/CN-10.0. From wavelet transform (WT) of $\chi(k)$ spectra (Figure 2d; and Figure S20, Supporting Information), we found that the centers of $[\chi(k), \chi(R)]$ intensity maximum for Cu/CN-10.0 are closer to that of Cu–N bond. Moreover, the best

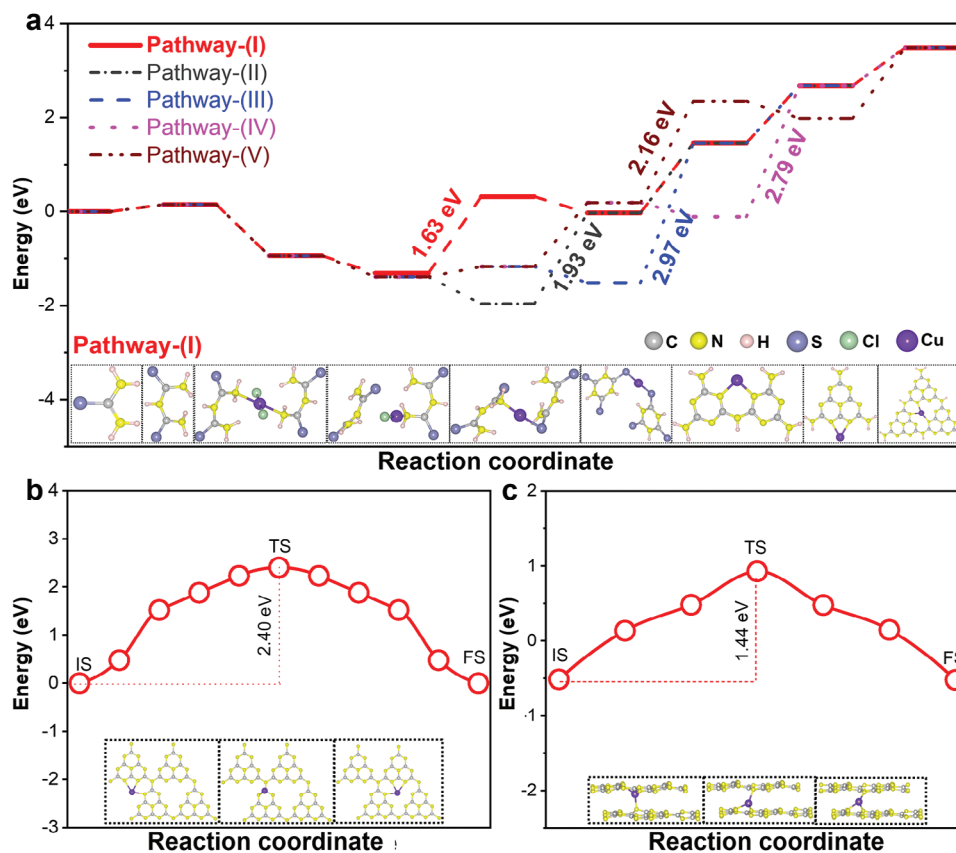


Figure 3. a) Calculated energy profile for stepwise formation of Cu/CN SAC with insets showing the optimized structures (Pathway-I). b,c.) Diffusion of a Cu single atom on the basal plane b) and in the interlayer c), respectively. The optimized initial (IS), transition (TS), and final states (FS) are shown in insets. Gray, yellow, pink, dark blue, purple, and green balls represent C, N, H, S, Cu, and Cl atoms, respectively.

FT-EXAFS fitting results of R space (Figure 2e) and k space (Figure S21, Supporting Information) show that the Cu centers in Cu/CN-10.0 are bonded with three N atoms. These results confirm the Cu atoms are isolated from each other and bonded only with three N atoms, in excellent agreement with the HAADF-STEM result. Furthermore, the O 1s XPS spectra (Figure S22, Supporting Information) show no peaks assigned to the Cu–O (530.4 eV) but only peaks corresponding to surface adsorbed O_2 , excluding the possible oxidation of Cu in air atmosphere. In addition, electron paramagnetic resonance (EPR) spectra (Figure S23, Supporting Information) reveal three absorption-like peaks of Cu, implying that the Cu center is coordinated by three N atoms. Such feature is attributed to the hyperfine interactions between the Cu atom and adjacent N nuclei.^[36,37] Based on the above experimental evidence, the local structure of Cu single atoms in Cu/CN-10.0 as a Cu- N_3 coordination environment is proposed in Figure 2f.

2.3. Mechanistic Investigation of the Synthesis of Cu/CN SAC

In order to get a deep insight into the advantages of our method, first-principles calculations are performed to study the elementary reaction steps involved in the synthetic process. The whole synthesis pathway proceeds initially from the complexation

between $CuCl_2$ and amide groups of biuret, which is frequently utilized to extract metal ions from industrial effluent.^[38] Then, there are only five possible pathways to form SACs depending on when the chloride ligands are removed. Specifically, pathway I has the two chloride ligands removed before the polymerization of s -triazine, whereas the removal of the two chloride ligands occurs after the s -triazine polymerization in the other four pathways (Figure S24, Supporting Information). As shown in Figure 3a, pathway I is the least endothermic, so our synthesis would proceed via this pathway, reflecting an in situ transformation of Cu/CN from the formed complexes. Particularly, the complexation process is remarkably exothermic, confirming the ease of capturing single-atomic Cu species. All the rest steps except for the polymerization of s -triazine are endothermic. These results are consistent with our experimental procedures, in which the complexation occurs at room temperature and only one calcining step (so-called the one-step method) is needed to produce Cu/CN. Comparatively, in most syntheses of CN-based SACs, metal atoms are postintroduced after CN substrates are formed.^[27] Calculations about this route (Figure S25, Supporting Information) reveal that both the production of CN and subsequent Cu/CN formation from $CuCl_1/CN$ are endothermic, so two calcining steps are required (so-called the two-step method), agreeing well with most reported preparation processes for CN-based SACs.^[15,21,26,27] In general, more heating steps would result

in larger-amount loss of SAC samples. As the in situ transformation feature in our strategy needs only one heating step, large amount of the SAC mass can be preserved, being responsible for the high mass yield (Table S3, Supporting Information).

In fact, the final polymerization steps from *s*-triazine (Figure 3a) are the well-documented procedures of synthesizing CN,^[15] indicating that the capture of Cu single species occurs prior to the formation of CN in our approach. This is in sharp contrast to the synthesis process of two-step methods, where Cu single atoms are incorporated after CN forms. In order to evaluate the stability of Cu/CN from the kinetic point of view, the diffusion behaviors of Cu single atoms on the basal plane and in the interlayer of CN are calculated, as shown in Figure 3b,c, respectively. It shows that the two processes need to overcome remarkably large barriers, reflecting rather difficult diffusion and thus high stability of the trapped Cu single metal atoms. According to our previous study,^[39] other 3d transition metal single atoms also reveal high stability once they are trapped in the cavity of CN. The high stability of the trapped single metal atoms and the exothermic nature of anchoring single metal species collectively confirm that single atoms energetically prefer the dispersive distribution. Combined with the capture of single metal species prior to CN formation and the complexation between CuCl₂ and biuret that could maximize the probability of bonding Cu cations to all available coordination sites, our method effectively protects single atoms from aggregation, accounting for the ultrahigh density loading.

2.4. Versatility of M/CN SACs and their Photocatalytic Performance

In order to assess the photocatalytic properties of the as-synthesized Cu/CN SACs, the photo absorption behavior is studied by UV-vis absorption (Figure S26, Supporting Information). Notably, the incorporation of Cu single atoms results in remarkable redshift of the absorption edge, reflecting an obviously improved light harvest.^[14] Besides, the SAC samples exhibit much weaker photoluminescence intensity than the pristine CN (Figure S27, Supporting Information), indicating dramatically suppressed recombination of photogenerated carriers.^[40] Furthermore, first-principles calculations reveal remarkable charge distribution in Cu/CN (Figure S28, Supporting Information), demonstrating a strong bonding between the Cu atom and the CN substrate. As a result, photogenerated electrons can be fast transferred via the strong bonds, facilitating the separation of photoinduced carriers. Impressively, increasing the mass loading boosts the light harvest and remarkably facilitates the separation of photogenerated carriers.

Motivated by above favorable attributes induced by introducing Cu single atoms and particularly the high mass loading, CO₂ photoreduction is investigated as a prototypical probe reaction (Figures S29–S31, Supporting Information). This selection is due to its great potential in alleviating the global climate change and energy shortage.^[41–43] As shown in Figure 4a, pristine CN reveals extremely low performance. Once Cu single atoms were introduced, abundant HCOOH and trace amount of CH₄/CO are yielded over Cu/CN, indicating the great promotion effects of Cu single atoms. Theoretical simulations indicate that the

reaction path of HCOOH formation over Cu/CN is energetically more favorable than that of CO production (Figure S32, Supporting Information), being responsible for the predominant yield of HCOOH. Additionally, in situ DRIFTS spectra combined with theoretical calculations suggest that the conversion from *COO to *COOH becomes more facile upon the introduction of Cu single atoms (Figures S32–S33, Supporting Information). Particularly, increasing the amount of Cu single atom loading monotonously enhances the photo(electro)chemical properties and eventually the photocatalytic performance of CO₂ reduction. All these results clearly manifest the promotion effect of Cu single atoms. Particularly, increasing the amount of metal loading monotonously enhances the photocatalytic performance. These results clearly manifest the promotion effect of Cu single atoms on the photocatalytic properties. More importantly, the observed loading-dependent activity reflects that larger mass loading yields higher surface density of active centers to catalyze CO₂ photoreduction.

After light irradiation for 4 h, the highest yield of HCOOH reaches 274.88 μmol g^{−1} h^{−1} for Cu/CN-10.0 (Figure 4a), which has also been confirmed in experiments conducted on both samples via small-scale and large-scale syntheses (Figure S34, Supporting Information). The product and electron selectivity of CO₂ reduction to HCOOH is calculated to be 92.01% and 89.14%, respectively (Figure S35, Supporting Information). Such a value is among the best reported values in existing C₃N₄-based catalysts (Table S7, Supporting Information). The remarkably high performance observed on Cu/CN-10.0 is primarily attributed to its exceptional charge carrier dynamics (Figures S36, S37 and Table S8, Supporting Information), high intrinsic activity^[44] (Figures S38 and S39, Supporting Information) and less possibility of by-products formation (Figure S40, Supporting Information). Note that, the formation of Cu clusters results in negative effects in the three aspects mentioned above and thus dramatically decreases the catalytic performance when the metal loading rate reaches 12.5 wt%. In addition, by altering the atmospheric treatment condition within the proposed strategy, a Cu-N₂ coordination counterpart can be also achieved (Figures S41, S42 and Table S9, Supporting Information), and its photocatalytic activity is found to be significantly lower than that of the Cu-N₃ coordination environment (Figure S43, Supporting Information), which may be due to its inferior efficiency of charge separation and transfer (Figure S44, Supporting Information). In addition, control experiments in the absence of CO₂, H₂O, photocatalyst, or light irradiation (Figure S45, Supporting Information) revealed that no products were detected. This result provides strong evidence that CO₂ is the only carbon source of HCOOH and CH₄/CO products. As durability is another crucial factor for commercial application, we then tested the durability of Cu/CN-10.0, as shown in Figure S46 (Supporting Information). Clearly, the product yield shows negligible decreases during a 5-run cycling test, TEM (Figure S47, Supporting Information) and AC HAADF-STEM (Figure S48, Supporting Information) characterizations about Cu/CN-10.0 after cycling test reveal that a robust stability of the prepared SAC samples, even at such high mass loading.

Finally, this metal coordination route is generalized to synthesize a series of transition metal single atoms M/CN (M = Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, and Zn) SACs with a metal loading of 0.50 wt%. As shown in Figure S49, Supporting Information, the

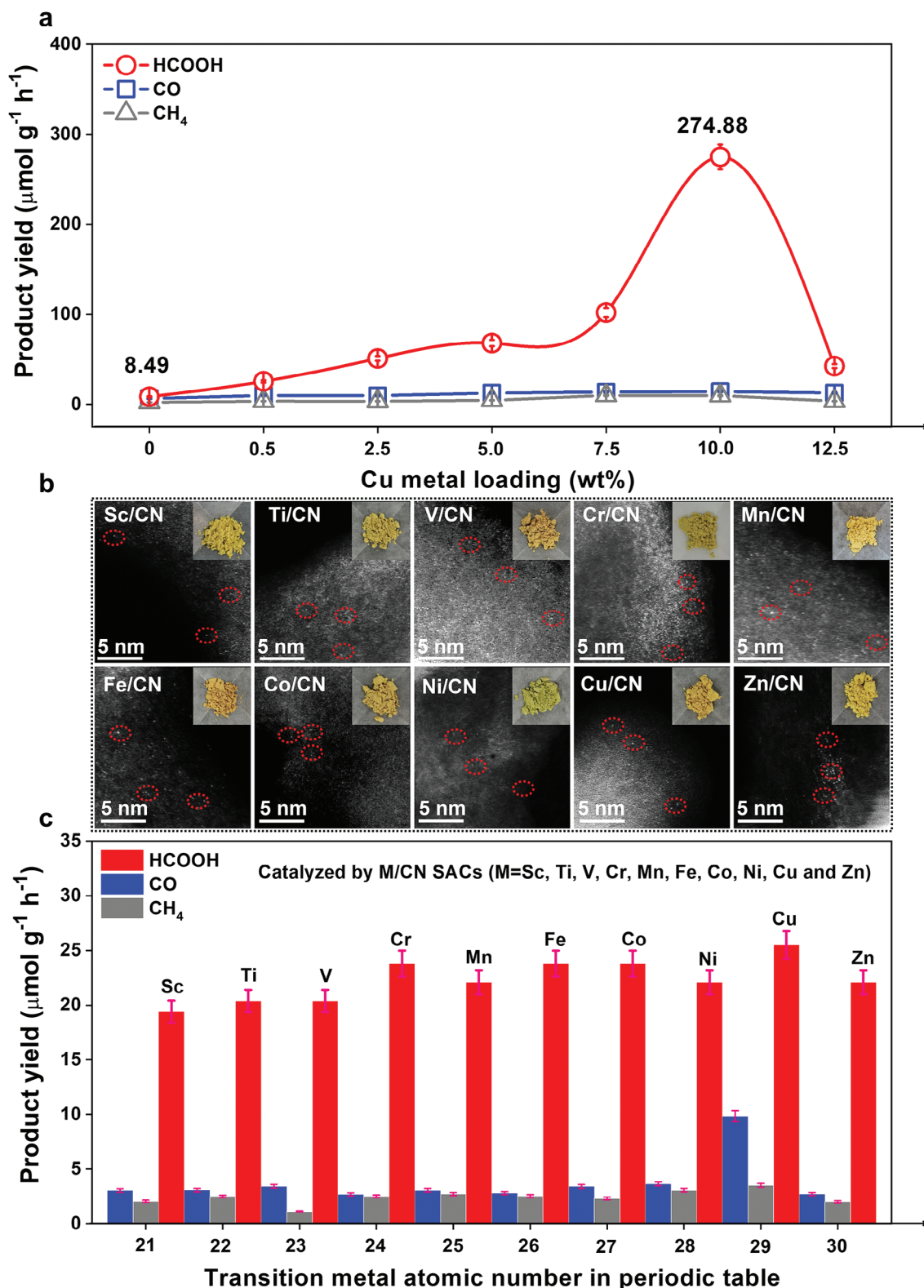


Figure 4. a) Products (HCOOH/CO/CH₄) of CO₂ photocatalytic reduction over CN and Cu/CN-x samples. b) AC HAADF-STEM images of M/CN (M = Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, and Zn) with a metal loading of 0.50 wt%. Insets show the digital photos of corresponding M/CN SACs samples. Single atoms are highlighted by red circles. c) Photocatalytic performance of transition metal SACs for CO₂ photocatalytic reduction. Reaction condition: 5 mg catalyst, solution containing Na₂S·9H₂O (0.35 mol L⁻¹) and Na₂SO₃ (0.25 mol L⁻¹) for CO₂ photoreduction. Error bars represent the standard errors of three independent measurements using 5 mg fresh sample for each measurement.

XRD patterns of all M/CN SACs present two peaks at around 13° and 27° , which are the typical diffraction peaks of CN. No peaks corresponding to transition metal-based compound can be detected, suggesting the high dispersion of transition metal species of M/CN. In addition, AC HAADF-STEM is used to confirm the atomic dispersion of these transition metal atoms. One can clearly see the presence of bright dots (indicated by red circles) in the AC HAADF-STEM images (Figure 4b), indicating that these metal atoms exist in mono-dispersed form. It is noted that there are no observable clusters or nanoparticles in the AC HAADF-STEM images. These results clearly demonstrate that this metal coordination route is a general approach for the synthesis of a variety of transition metal SACs, which are very difficult to be achieved via conventional pyrolysis methods. Further, CO_2 reduction tests are carried out to probe the photocatalytic performance of the prepared M/CN SACs. As shown in Figure 4c, the main products are all HCOOH and, specifically, they also possess remarkably higher activity than the pristine CN ($8.49 \mu\text{mol g}^{-1} \text{h}^{-1}$). These findings not only provide impressive candidates for CO_2 photoreduction, but also represent a paradigm of experimentally screening advanced materials.

In order to gain deep insight into the activity enhancement as well as the activity difference of the ten M/CN SACs, UV-vis absorption, photoluminescence (PL), and time-resolved photoluminescence (TRPL) tests were conducted. As shown in Figures S50, S51 and Table S10 (Supporting Information), though different M/CN SACs behave differently, the introduction of single atoms dramatically enhances light absorption and promotes charge carrier separation and transport of CN. Moreover, in order to further understand the enhanced activity and particularly identify the nature of the active center of M/CN SACs, theoretical calculations were performed about the stability of the key reaction intermediate $^*\text{COOH}$. Four nonequivalent sites were considered, including three N sites around the cavity of CN and the metal center (Figure S52, Supporting Information). As shown in Table S11 (Supporting Information), $^*\text{COOH}$ at the metal site exhibits significantly lower total energy on all M/CN SACs, indicating that the metal sites should be the active centers for CO_2 reduction. Subsequently, the elementary steps ($\text{CO}_2 \rightarrow ^*\text{COOH} \rightarrow ^*\text{HCOOH} \rightarrow \text{HCOOH}$) involved in CO_2 reduction to HCOOH were calculated. A comparison of the Gibbs free energy profiles (Figure S53, Supporting Information) suggests that CO_2 reduction over M/CN is energetically smoother than that on pristine CN with the most uphill transformation steps being less endothermic by 21%–47% (Figure S54, Supporting Information). These calculated results indicate that the incorporated single metal atoms serve as the dominant active centers to facilitate the conversion from $^*\text{COO}$ to $^*\text{COOH}$, and ultimately boost the yield of HCOOH.

3. Conclusions

In summary, a metal coordination route is proposed and demonstrated to achieve industrial-scale synthesis of a series of transition metal SACs (M/CN, M = Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, and Zn) in air. The strength of this method lies in the rapid and firm complexation between metal ions and biuret and the avoidance of multiple-step washing and heating, which contribute to largely preserved mass and in turn one-pot large-scale production. Im-

portantly, the strategy maximizes the probability of bonding between the metal and available coordination sites in amide groups, thereby leading to a high-density loading. Specifically, a massive production of up to 3.565 kg and a high metal loading of 16.03 mol% for Cu/CN are obtained. The as-prepared Cu/CN SACs exhibit outstanding activity, selectivity and durability for CO_2 photoreduction to HCOOH. Particularly, increasing the amount of metal loading enhances the photocatalytic performance gradually, indicating that larger metal loading provides higher surface density of active centers. The proposed facile and versatile strategy not only provides an ideal platform for experimentally screening advanced catalytic materials but also is ready for further scale-up which paves the way for the industrialization of SACs.

4. Experimental Section

Sample Preparation—Materials: Thiourea ($\text{CH}_4\text{N}_2\text{S}$, Aladdin, 99%), scandium chloride (ScCl_3 , Aladdin, 99%), titanium (IV) chloride solution (TiCl_4 , Aladdin, 99%), vanadium trichloride (VCl_3 , Aladdin, 97%), chromium chloride ($\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$, Aladdin, 98%), manganese chloride (MnCl_2 , Aladdin, 99%), ferric chloride ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$, Virtue-Clara, 99.5%), cobalt chloride (CoCl_2 , Aladdin, 99.7%), nickel chloride ($\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$, Aladdin, 98%), copper chloride ($\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$, Aladdin, 99%), copper sulfate (CuSO_4 , Aladdin, 99%), copper nitrate ($\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$, Aladdin, 99%), copper acetate ($\text{Cu}(\text{CO}_2\text{CH}_3)_2 \cdot \text{H}_2\text{O}$, Sigma-Aldrich, 99%), zinc chloride (ZnCl_2 , Aladdin, 99.95%). All chemicals used in this study were analytical-grade reagents without further purification. Deionized water was utilized in all experiments.

Sample Preparation—The synthesis of M/CN: In a typical synthesis of Cu/CN, 10 g thiourea was dissolved in 30 mL deionized water to form a clear transparent solution by ultrasound treatment. A small amount of $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ was added slowly into the solution and then the mixture was magnetically stirred for 3 h. Subsequently, white crystals were collected by centrifugation, followed by drying at 343 K for 3 h in an oven. The white crystals were heated to 823 K at a rate of 10 K min^{-1} in air and maintained at this temperature for 3 h. Then, the obtained sample was placed in a planetary ball mill (FRITSCH, Pulverisette 7 premium) and milled for 0.5 h at 400 rpm. The Cu/CN specimens with different Cu loading were labeled as Cu/CN- x , where $x = 0.5, 2.5, 5.0, 7.5, 10.0$, and $12.5 \text{ wt\%}$. Other Cu salts, including CuSO_4 , $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$, and $\text{Cu}(\text{CO}_2\text{CH}_3)_2 \cdot \text{H}_2\text{O}$, were also utilized to synthesize Cu/CN-10.0. Besides, M/CN (M = Sc, Ti, V, Cr, Mn, Fe, Co, Ni, and Zn) were also synthesized using thiourea and corresponding metal chloride according to the same procedures. Bulk CN was also prepared as a reference sample by heating the thiourea according to identical procedures.

Material Characterization: X-ray diffraction (XRD, PANalytical B.V. X'Pert Powder) with Cu $K\alpha$ radiation was employed to reveal the crystal structure of as-obtained specimens (at 40 kV and 40 mA). Aberration corrected high-angle annular dark-field scanning transmission electron microscopy (AC HAADF-STEM) and corresponding energy-dispersive X-ray spectroscopy (EDS) images were collected on a FEI Titan Themis at 300 kV. Electron paramagnetic resonance spectroscopy spectra were collected on Chinainstru&Quantumtech (Hefei) EPR200-Plus. The Ultraviolet-visible (UV-vis) diffuse-reflectance spectrometry (DRS) spectra were obtained for the dry-pressed disk samples using a UV-vis spectrophotometer (Shimadzu UV-3600) equipped with an integrating sphere assembly, using 100% BaSO_4 as the reflectance sample. A steady state fluorescence spectrometer (Shimadzu RF-6000) was utilized to get the photoluminescence spectra and the excitation wavelength was 375 nm. The actual chemical compositions of the catalysts were measured by inductively coupled plasma-optical emission spectrometer (ThermoFisher Scientific, ICP 6300 DUO).

Photocatalytic CO_2 Reduction: 5 mg photocatalysts were ultrasonically dispersed in 5 mL solution containing $\text{Na}_2\text{S} \cdot 9\text{H}_2\text{O}$ (0.35 mol L^{-1}) and Na_2SO_3 (0.25 mol L^{-1}) in a quartz beaker. Then the beaker was placed in a 150 mL reaction chamber (Beijing Perfectlight Technology Co., Ltd.

PLR MFPR-I). The pressure was maintained at 105 kPa. Finally, the reactor was placed under a 300 W xenon source (Beijing Perfectlight Technology Co., Ltd. PLS-SXE300/300UV) to initiate CO₂ photoreduction and the temperature of the solutions was controlled at 293 ± 0.2 K by a recirculating cooling water system (DC-0506, Geng gong Instrument, Shanghai) during irradiation. Every 1 h, the autosampler injected the quantitative gas in the reactor into gas chromatography (GC-2014C, SHIMADZU) to perform product analysis (H₂, CO, CH₄). The product in the liquid phase was identified by ¹H NMR spectroscopy (Avance NEO 400, Bruker). The product yield was calculated from the integration of ¹H NMR peaks using dimethylsulfoxide (DMSO) and D₂O as the internal standard.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

Keywords

CO₂ photoreduction, large scale preparation, metal coordination route, single atom catalysts, ultrahigh metal loading

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